D4GC 2026

FieldMed

Offline answers for everyone on an MSF field team

Ask in your language and get the right document with its source, even with no internet.

Doctors and nurses | Logisticians | Water and sanitation | Coordinators | Admin and HR

setting up a cholera isolation ward

The problem

Every role on a team needs answers it can trust, offline

MSF works in 70+ countries, often with poor internet, little storage and old laptops. The questions come from everyone.

A nurse

"How do I manage bleeding after childbirth?"

A logistician

"How do we lay out an isolation ward for cholera next to measles?"

A coordinator

"What has MSF done on cholera in DR Congo before?"

Wrong answers cost

An invented dose, or a bad site layout, puts patients and staff at risk.

Knowledge leaves with people

What a team learned in North Kivu rarely reaches the next team there.

Gaps stay invisible

Nobody records which questions go unanswered, or where.

Our solution

One laptop serves the whole site

Phones only need a browser. The laptop runs everything locally and syncs when a connection appears.

Off-site build

Guidelines, reports, briefs and field stories are cleaned, deduplicated, chunked, tagged by country and topic, and indexed in SQLite with embeddings.

On site, fully offline

One ordinary laptop answers every phone and tablet on the local Wi-Fi. No internet needed.

Central hub, when online

Pulls new protocols down. Sends field experiences and knowledge gaps up.

100% open source

CPU only

No vendor lock-in

€0 per query

How a question is answered

Evidence first, summary second

Every step runs on the laptop's CPU.

1 · Ask

Type or speak. Speech-to-text runs offline with Whisper.

2 · Search two ways

Keywords (BM25) and meaning (multilingual-e5-small), fused into one list.

3 · Rerank

A cross-encoder scores each passage. Below the threshold, nothing is shown.

4 · Sources, then summary

A local Qwen3 0.6B writes a short summary and must cite [1], [2]. Bad citations are flagged.

5 · Explain

Confidence score and "Why this answer?": relevance per source, how it was found, matched words.

6 · Read aloud

The device's offline voice reads the answer when hands are busy.

Results, measured on a laptop CPU

Each layer helps. Only the reranker can say "I don't know"

48 test questions: 23 written from titles, 20 from inside documents, 5 with no answer in the collection. Ryzen 7 3700U, no GPU.

Setup	Hit@1	Hit@5	MRR	Off-topic rejected	Time
Keyword only	0.86	0.95	0.90	0 / 5	12 ms
Meaning only	0.81	0.98	0.88	0 / 5	37 ms
Combined	0.91	0.95	0.93	0 / 5	55 ms
Combined + reranker	**0.95**	**0.98**	**0.96**	**5 / 5**	**1.9 s**

562 documents and 18,355 passages fit in a 36 MB index plus 14 MB of embeddings.

Beyond search

Built for how field teams work

French, Spanish, Arabic

Questions are translated, searched in English, and FR/ES answers come back translated, with the original shown.

Share field experience

Any role can log what worked and what failed. Pseudonymous, region-level location, marked unverified until approved.

Map and wiki

Browse documents and experiences by country and topic, offline.

Knowledge gaps

Unanswered questions are logged by area and topic. Three or more flag the area for new material.

Region packs

Going to DR Congo? Take 130 documents plus all guidelines: about 10 MB instead of 50.

Hub sync

Every two minutes when online. Answers come from the hub if reachable, from the laptop if not.

Follow-ups like "and for children?" keep the context of the last three turns.

Safety by design

No answer is better than a wrong answer

Sources before the summary

The model only sees reranked passages and must cite them. Invalid citations are flagged on screen.

A relevance threshold

Passages under 12% relevance are dropped. If nothing is left, the app says so.

Doses copied word for word

A banner on every summary: check the cited source before acting.

Staff and patient privacy

Pseudonyms, region-level location, an identifier check. Data stays on the device.

Asked: "best restaurants in Brussels"

No answer · 0%

No relevant document found. Try other words or remove filters.

The question is logged as a knowledge gap.

Cost and implementability

Runs on the laptop the project already has

Per query
€0

GPU needed
None

Tested RAM
14 GB

Region pack
~10 MB

Open-source parts only

SQLite, FastAPI, e5, MiniLM, Whisper, Qwen3 (Apache 2.0) and Leaflet. No cloud API, no licence fees.

Any document, any department

Adding a logistics manual or an HR policy means rerunning the build script. A better local model swaps in with one config line.

Roadmap

What we'd build next

Better answers

Fine-tune embeddings and reranker on MSF question-answer pairs, a small LoRA-tuned model on MSF documents, and confidence calibrated on a labelled set.

Phones without the laptop

Run Qwen3 0.6B in the phone's browser (WebLLM or wllama).

Full multilingual voice

Multilingual Whisper, Arabic answers from a larger model, French and Arabic tagging.

Trust and review

An approval workflow for team leads, encrypted packs, authentication.

Photo-assisted triage

Wound, burn or fracture photo matched to the right protocol. Opt-in, with consent, no faces.

Ask a colleague

Reach staff who worked on the same project or topic, from an opt-in directory. Plus OCR for scans and sync over SMS or satellite.

The ask

FieldMed gives every person on a field team the right document, shows its evidence, says when it doesn't know, and keeps each team's knowledge for the next one.

Let us pilot it in one MSF project, across all roles.

We need an ordinary laptop on site, the project's region pack, the documents each department relies on, and someone to review shared experiences.

Appendix

What doesn't work yet

We'd rather you hear these from us.

The test set is ours

48 questions written by the team, so results are likely optimistic. One known miss: for "standard TB regimens", annual reports outrank the guideline.

Confidence isn't calibrated

High relevance means the topic matches, not that the source is authoritative.

Small model, basic summaries

Qwen3 0.6B. Arabic answers stay in English, translation quality is unmeasured, and voice input is English only.

Prototype hub

A local stand-in without authentication. Field experiences don't yet rank below official guidance; the unverified label is the safeguard.